

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

MSE 2201

(Phase Diagrams and Transformations)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What is Non-equilibrium cooling? How composition differs in case of non-equilibrium cooling from equilibrium cooling? Which one is more preferable to get uniform mechanical properties and why? (15)
- (b) For a 40 wt% Sn - 60 wt% Pb alloy at 150 °C, calculate the relative amount of each phase present in terms of (a) mass fraction, and (b) volume fraction. At 150 °C take the densities of Pb and Sn to be 11.23 and 7.24 gm/cm³, respectively. Fig. 1 (b) phase diagram of Lead-tin alloy attached. (15)

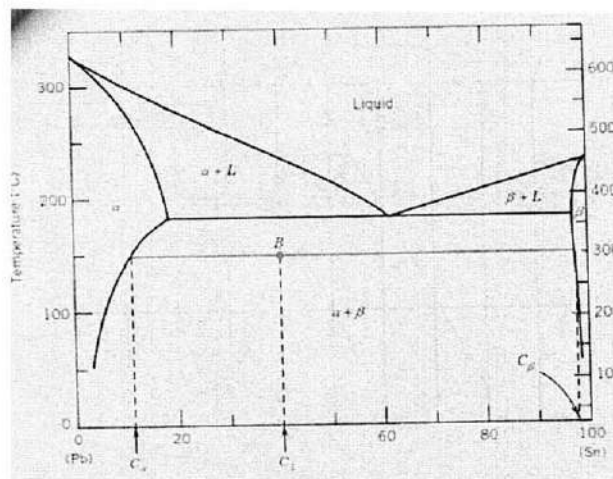


Figure 1 (b)

- Q 2. (a) Draw and explain the schematic representations of the microstructures for an iron-carbon alloy of hypoeutectoid compositions, as it is cooled from within the austenite phase region to below the eutectoid temperature. (20)
- (b) What is allotrope? How iron shows allotropic properties? Explain it with cooling curve of pure iron. (10)
- Q 3. (a) Explain the schematic representations of the microstructure for an iron-carbon alloy of hypereutectoid compositions, as it is cooled from austenite phase region to below the eutectic temperature. (20)
- (b) What do you understand by Degree of freedom as 2. (05)
- (c) Why peritectic reaction is so called an incomplete reaction? (05)

Section B

(Answer ANY TWO questions from this section in **answer script B**)

- Q 4. (a) Name three methods of constructing phase diagrams. How will you construct a phase diagram based on the resistivity of the material? (15)
- (b) Identify the intermediate phases in the given phase diagram. Which of these are intermetallics? How do you differentiate intermetallics from intermediate phases? (15)

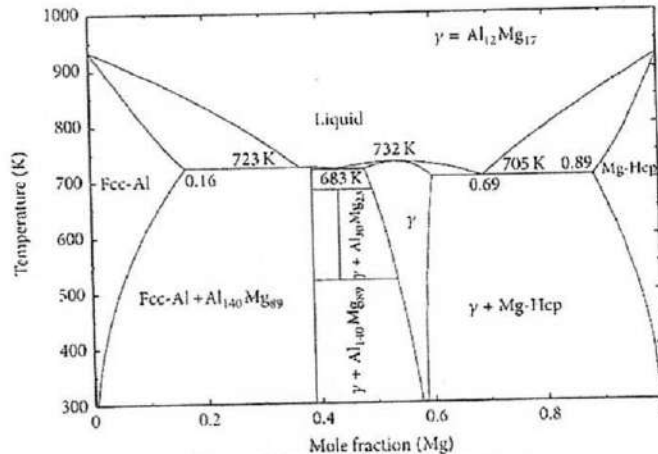


Figure 4 (b) : Al-Mg Phase diagram

- Q 5. (a) What is the necessity of ternary phase diagrams? Can an isothermal ternary diagram completely define a ternary phase transformation? Justify your answer. (15)
- (b) Answer using the Silica ternary phase diagram in figure 5 (b). (15)

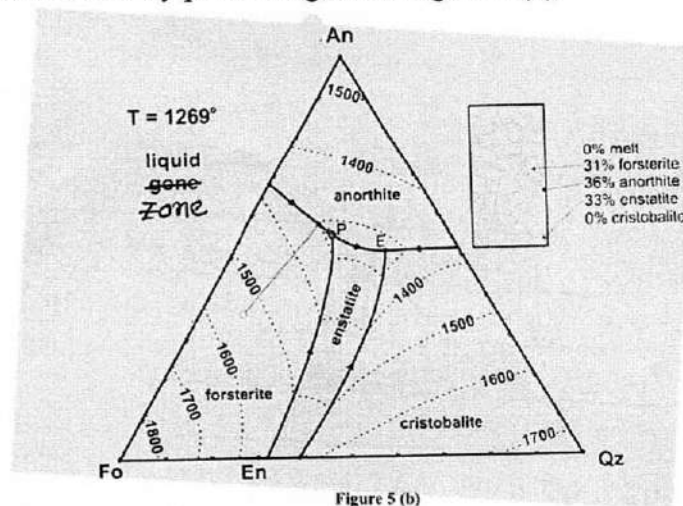


Figure 5 (b)

What are the final products of cooling Forsterite and enstatite at different cooling rates (Use hashed lines, solid lines or, different colored pens/pencils for marking the cooling rates).

- Q 6. (a) What do you understand by supercooling? Draw the common microstructure that forms due to supercooling of liquid aluminium. (12)
- (b) Why does the composition gradient occur during solidification? Explain. (08)
- (c) 'Phase diagram is an important tool for materials design'- convey your remarks on this statement. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

MSE 2203

(Mechanical Behavior of Materials)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What will happen if a strain-hardened material undergoes annealing? (12)
(b) How does True stress-strain curve differ from Engineering stress-strain curve? (08)
(c) In a tensile test, two points on a stress-strain curve for a material are $\sigma = 165$ MPa at $\epsilon = 0.06$ and $\sigma = 296$ MPa at $\epsilon = 0.14$ respectively. Find the strain hardening exponent and strength coefficient of the material and predict σ at $\epsilon = 0.20$. [Assume Power Law for calculations]. (10)
- Q 2. (a) Discuss the strain fields surrounding an edge dislocation and their interaction with each other. (10)
(b) Calculate the activation energy for vacancy formation in aluminum, given that the equilibrium number of vacancies at 645°C is $8.68 \times 10^{23} \text{ m}^{-3}$. The atomic weight and density (at 645°C) for Al are 26.98 g/mol and 2.62 g/cc respectively. (08)
(c) How can crystal orientation in case of resolved shear stress make it easy or hard to move dislocations? (12)
- Q 3. (a) Consider a single crystal of BCC iron oriented such that a tensile stress 65 MPa is applied along [010] direction. The critical resolved shear stress is 40 MPa for (110) $[\bar{1}11]$ slip system. Is there any slip? If no slip occur, what will happen finally to material with increasing load? (12)
(b) What are the differences between climb and slip? (06)
(c) The stress components in a part of a structure have been calculated to $\sigma_x = 112$ MPa, $\sigma_y = 75$ MPa and $\tau_{xy} = 54$ MPa (all the other stress components are zero). For this structure a safety factor $S = 1.45$ with respect to yielding is required. What yield strength (σ_y) should the materials have to fulfill this condition? Investigate both the Tresca and von Mises yield criteria. (12)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) How does ductile failure occur? Explain with necessary figures. (10)
- (b) Is Griffith's theory applicable to brittle metal? Justify your answer. (08)
- (c) Determine the theoretical and actual fracture strength (considering the crack) of a silica fiber, (12)
if $E = 67 \text{ GPa}$, $\gamma_s = 1.4 \text{ J/m}^2$, $a_0 = 0.13 \text{ nm}$, and crack length $(c) = 10^5 a_0$.
- Q 5. (a) "A notch in a thin plate is far more damaging than in a thick plate"- is this statement true? (14)
Justify your answer with the necessary figures.
- (b) A large plate is fabricated from a steel alloy that has a plane strain fracture toughness of $82.4 \text{ MPa } \sqrt{\text{m}}$. (10)
If, during service use, the plate is exposed to a tensile stress of 345 MPa , determine the minimum length of a surface crack that will lead to fracture. Assume a value of 1.0 for α .
- (c) How does thermal shock differ from thermal fatigue? (06)
- Q 6. (a) How do surface properties affect fatigue behaviors of metals? (10)
- (b) Draw a suitable fracture mechanism map and explain how fracture occurs at elevated (12)
temperatures.
- (c) Superimpose on the same strain-vs-time plot schematic creep curves for both constant tensile (08)
stress and constant tensile load, and explain the differences in behavior.

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Math 2227
(Statistics, Fourier Series and Numerical Analysis)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Define raw moments and central moments. If the first four moments of a distribution about the value 5 are equal to -4, 22, -117 and 560. Determine the corresponding moments about the mean. Hence find β_1 and β_2 . (16)
- (b) What do you mean by conditional probability? From a group of 10 children (6 boys and 4 girls), 3 children are selected at random. Calculate the probabilities that selected group contains (i) no boys, (ii) at least one girl, and (iii) more girls than boys. (14)
- Q 2. (a) A sample of 100 dry cells tested to find the length of life produced the following results: $\bar{x} = 12$ hours, $\sigma = 3$ hours. Assuming the data to be normally distributed. What is the percentage of battery cells are expected to have life (14)
- (i) more than 15 hours; and (ii) between 10 and 14 hours.
- (b) Find the co-efficient of co-relation between temperature and strength of bonding of a solid material. The experimental data are given below: (16)

Temperature	15	20	30	40
Strength	7	11	20	32

- Q 3. (a) Write down Dirichlet's condition for Fourier series. Obtain the Fourier series expansion of the function $f(x) = x$, $0 < x < 2\pi$. (18)
- (b) Find half range Fourier cosine series of the function $f(x) = \sin 2x$, $0 < x < \pi$. (12)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Evaluate the following (the interval of differencing being unity). (05)

$$\Delta(x+9)!$$

- (b) Find the number of men getting wages per hour between Tk. 10 and Tk. 15 from the following table. (15)

Wages in Tk.	0-10	10-20	20-30	30-40
Number of men	9	30	35	42

- (c) Find the third divided difference with arguments 2, 4, 9, 10 of the function $f(x) = x^3 - x$. (10)

- Q 5. (a) Using Lagrange's interpolation formula, find the polynomial $f(x)$ which takes the following values: (12)

x	0	1	3	5
f(x)	-10	2	9	14

- (b) The tabulated values of a function $f(x)$ are given below. Find the first derivative of the function at $x = 50$. (08)

x	50	51	52
y = f(x)	3.6840	3.7084	3.7325

- (c) Evaluate $\int_0^6 \frac{dx}{1+x}$ upto three decimal places by dividing the range into six equal parts, using Simpson's $\frac{1}{3}$ rule. (10)

- Q 6. (a) Find a real root of the equation $x^2 + \ln x - 2 = 0$ in $[1,2]$ by Newton-Raphson's method, correct upto three decimal places. (15)

- (b) Solve the following system of linear equations by Gauss-Jacobi iteration method upto third iteration. (15)

$$x + 17y - 2z = 48; \quad 2x + 2y + 18z = 30; \quad 30x - 2y + 3z = 48.$$

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Ph 2227

(Basic Quantum Mechanics and Solid State Physics)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Describe the effect of potential on photoelectric current and the effect of frequency of incident light on stopping potential with proper diagram. (12)
- (b) Mention the quantum numbers associate with vector atom model. (10)
- (c) In a compton effect experiment in which the incident X-ray have wavelength of 11 pm, the scattered X-ray at a certain angle have a wavelength of 10.8 pm. Find the momentum of the corresponding recoil electrons (Assume magnitude and direction). (08)
- Q 2. (a) Define space quantization, spin quantization and Hermitian operator. Show that, the eigenvalue of a Hermitian operator is real. (09+11)
- (b) Two wave functions Ψ_1 and Ψ_2 are defined by following terms, (10)

$$\Psi_1 = \begin{pmatrix} -1 \\ i \end{pmatrix}, \Psi_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Show that, they are orthogonal in nature.

- Q 3. (a) What is wave function Ψ ? If an wave function is defined by, (12)

$$\Psi(x,t) = Ae^{i(Kx-\omega t)}$$

Then, (i) Obtain the operator corresponding to momentum ($\hat{p}_x$), and (ii) Show that, $[\hat{x}, \hat{p}_x] = i\hbar$

- (b) In the 1-D space, derive the time dependent Schrödinger's equation. (10)
- (c) If the orbital angular momentum is expressed as, $\vec{l} = \vec{r} \times \vec{p}$, then obtain all the angular momentum operators i.e., $\hat{l}_x, \hat{l}_y, \hat{l}_z$. (08)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What are phonons? Explain the Laws of conservation of energy and momentum in the case of inelastic scattering of a photon by a phonon. (12)
- (b) Calculate the lattice specific heat according to the Debye theory. (10)
- (c) Calculate the maximum phonon frequency generated by scattering of visible light of wavelength $\lambda = 5890 \text{ \AA}$. Given that velocity of sound in medium is $5.2 \times 10^5 \text{ cm/sec}$ and refractive index is 1.49. (08)
- Q 5. (a) Discuss the outstanding properties of metal. There are 2.6×10^{28} free electrons per cubic meter of sodium. Calculate the Fermi energy and Fermi velocity. (07+08)
- (b) What are risk assessment requires from nanomaterials? Mention and discuss the classification of nanomaterials. (15)
- Q 6. (a) Discuss the following terms- (10)
- (i) Population inversion
 - (ii) Spontaneous emission
- (b) How can you be able to achieve higher probability of stimulated emission as compared to that of spontaneous emission? Explain. (10)
- (c) A 1500 KW LASER emits light of 900 nm wavelength. Calculate the number of photons emitted by LASER every second. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2020
Department of Materials Science and Engineering

CSE 2227
(Programming Techniques)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Differentiate between interpreter and compiler. Give example (s) of four control sequences. (08)
(b) Why do we need variables in C? Write down the valid and invalid variable (s) from the following- (12)
(i) age_ * (ii) 1 year (iii) _name (iv) + batch (v) class1 (vi) B-section (vii) class one (viii) (local)
(c) Write a program in C to display the n terms of even natural number and their sum. (10)
- Q 2. (a) Explain entry-controlled loop and exit-controlled loop using proper flowchart and examples. (10)
(b) Explain dangling else problem with example. Write down the working procedure of switch statement with example. (12)
(c) Write a program in C to give an example of declaration of symbolic constant. (08)
- Q 3. (a) Write a program in C to show the type conversions in expressions. (07)
(b) Why should goto be avoided in C? Explain break and continue statement with example(s). (10)
(c) Two one-dimensional array A and B are sorted in ascending order. Write a program to merge them into a single sorted array C that contains every item from arrays A and B in ascending order. (13)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Material scientists work with diverse types of materials that have different melting points. (10)
We are going to design a class namely 'Material' to store the name and melting point of various materials. Now,
- i) Design a parameterized constructor whose role is to initialize the name and melting point of different materials.
 - ii) Create a function that calculates and displays the average melting point of two materials when objects of the 'Material' class are passed to it.
- (b) Is there any way to overload destructor of a class in C++? Justify your answer with proper example. (10)
- (c) Is it possible for a static member function of a specific class to be created with a 'const' qualifier in C++? Justify your answer with proper example. (10)
- Q 5. (a) The Dept. of MSE, KUET has decided to keep track of all 240 students enrolled in the department. Roll number, Name, E-mail, Mobile number, CGPA, and Year of enrollment will be considered while storing the students' information. (14)
- i) Write a function that will take the 'Roll number' as a parameter and print all the stored information of that student.
 - ii) Write a function that will take the 'Year of enrollment' as a parameter and print all the students who had enrolled on that particular year in the MSE dept.
- (b) As a memory management function, is it better to use malloc () or calloc ()? Explain. (08)
- (c) Explain the following built-in string functions of C- (i) strcat (), (ii) strcpy (). (08)
- Q 6. (a) What will be the output of the following C program? Justify your answer with proper explanation. (10)
- ```
int main() {
 char *mse = "MSE-KUET";
 printf ("%c\n", *&*mse);
 return 0;}
```
- (b) Write down some advantages of using Null pointer in C programming. (07)
- (c) Write a program in C that can find a word from a string. (13)